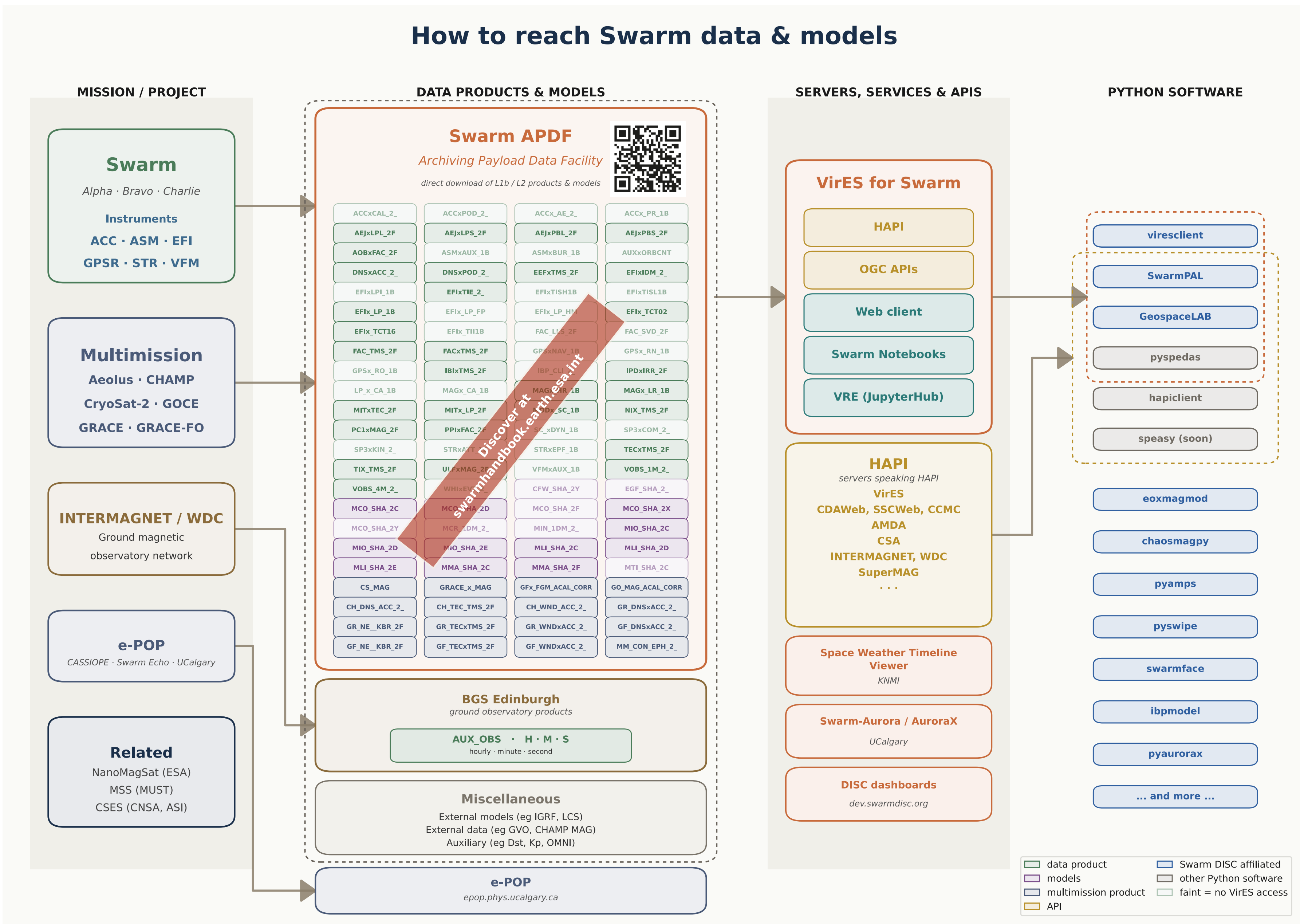


ESA Swarm products, tools, and services supporting open science

Ashley Smith (ashley@mag.earth) on behalf of Swarm DISC



How to reach Swarm data & models



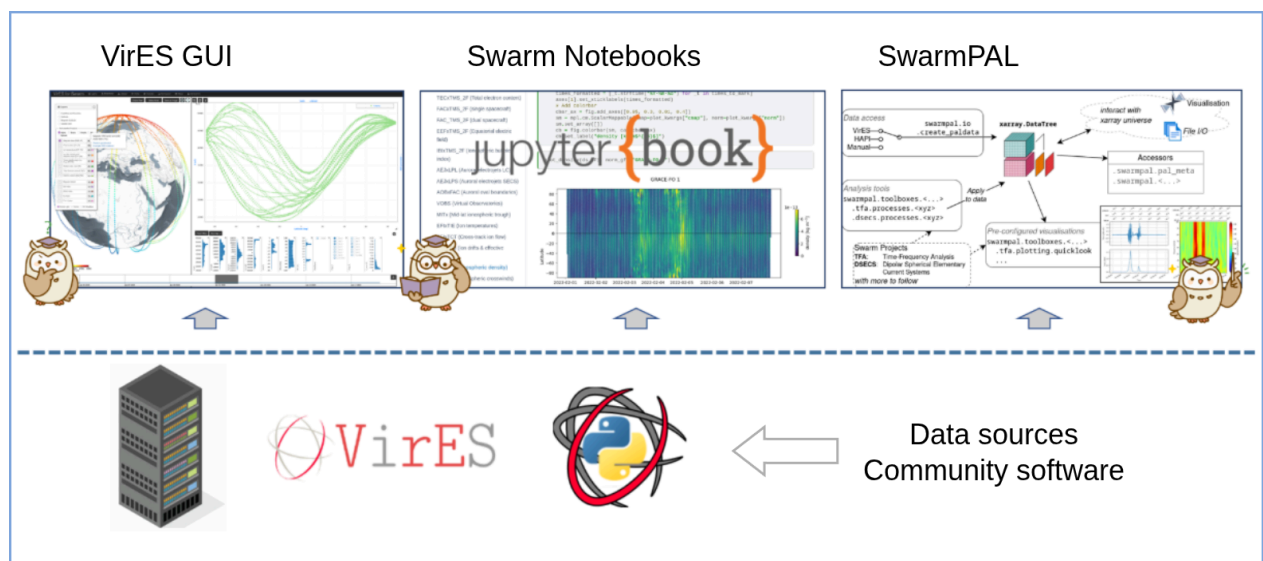
VirES infrastructure

Virtual environments for Earth Scientists

- **VirES web client:** Interactive 3D and 2D plotting
 - ▶ `vires.services`
- **VirES Server**
 - ▶ OGC WPS: Open Geospatial Consortium Web Processing Service
 - Full VirES API: access using `viresclient`
 - Operations include: on-demand model evaluation, data subsetting, time series aggregation, auxiliary data.
 - ▶ HAPI: Heliophysics API - Time series access.
 - Access via e.g. `hapiclient`. See `hapi-server.org` for details.

Python tooling:

- **viresclient:** Python package to interact with the server
 - ▶ `viresclient.readthedocs.io`
- **JupyterHub (VRE)** - just sign in; thematic community libraries provided
 - ▶ `vre.vires.services`
- **Swarm Notebooks** cookbook (ready to run in VRE)
 - ▶ `notebooks.vires.services`
 - ▶ Linked from product documentation in the Swarm handbook. Provides discoverability and quick access to data and software. Continuous integration supports maintenance as the software stack changes.
- **SwarmPAL:** Python package supporting analysis
 - ▶ `swarpal.readthedocs.io`
 - ▶ Modular toolboxes developed by researchers, using a shared input/output layer. Connects to VirES and any HAPI server for broad data access. Attaches software engineering expertise to improve extensibility and maintainability.



VirES services support both interactive exploration and development of portable research code, lowering the barrier to access Swarm products. The VRE is also used in educational contexts, allowing rapid onboarding to begin writing and running code.

- Open development:
 - ▶ `github.com/ESA-VirES`
 - ▶ `github.com/Swarm-DISC`

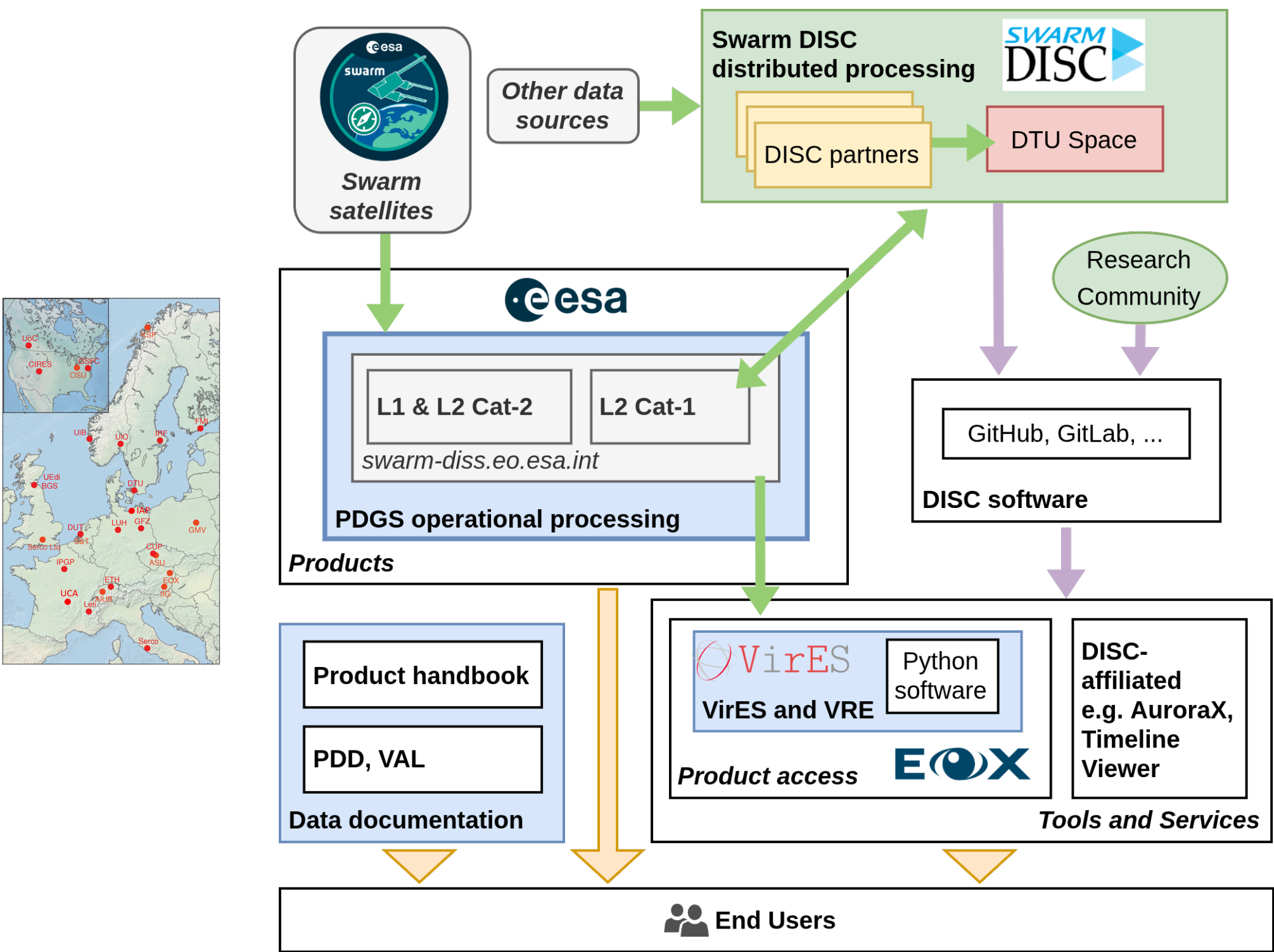
Swarm is comprised of three identical satellites in complementary polar orbits since 2013, providing magnetic and plasma measurements to probe both Earth's interior and geospace. The mission is augmented by so-called *multimission* products and increasing connectivity with other observation systems. Expanding the goals and capabilities of the mission, Swarm DISC supports the development of a range of software tools and services to empower and connect researchers and developers.

Swarm DISC

`swarmdisc.org`

The *Swarm Data, Innovation, and Science Cluster* is the consortium of institutions and companies that support the Swarm mission through the development and delivery of Swarm products, tools, and services.

While ESA is responsible for the Payload Data Ground Segment (PDGS) running the core data processing and dissemination, DISC extends this with distributed generation of higher level products.



How to sustainably grow the data/software ecosystem to accommodate increasing complexity?



VirES is funded by ESA and implemented by EOX IT Services GmbH.